PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
PRE – MOCK ASSESSMENT- FORM TWO
041 BASIC MATHEMATICS
TIME: 2.00 HOURS **March, 2025**

ANSWER ALL QUESTIONS

1. (a) Change 0.3636 … into fraction
(b) Find $4.2351 \div 3$ correct to four significant figures

2. (a) Find the values of $(2670)^2 - (670)^2$
(b) Write 56 as a product of a prime factors.

3. (a) If m*n = $(mn)^2$ find 2*3
(b) Express 0.0006457 in standard form

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4. (a) Rationalize the denominator $\frac{1}{3-\sqrt{3}}$
(b) Solve for x if $\frac{x+10}{x-4} = 3$

5. (a) Solve for x if $3x^2 - 7x + 2 = 0$
(b) Find the simple interest on 180,000/= for three years at the rate of 5% per year

6. (a) the side of m equilateral triangle is p, it is altitude is r. express r in terms of p if its area is A
b). solve $|6 - x| = 3$

7. (a) Find the equation of a straight line passing through the point (5 , 2) and (6 , 4)
b). find the intersection value of the following lines y = 2x + 1 and 2y – 3x – 1 = 0

8. (a) given the formula $T = 100n + \frac{k}{v^2}$
   i. Express v in term of T, n and k
   ii. Find the value of v if n = 25, k = 100, and T = 2750

   b). solve for n, if $\left(\frac{3}{5}\right)^{n-1} = \left(\frac{25}{9}\right)^{2n}$

9. A map is drawn to scale of 1:25,000. The length of the river on the map is 20cm what is actual length of the river in km?

10. a). The interior angle of a regular polygon is 120^0 greater than the exterior angle. Find the number of sides it has.
b). a father is now three times as old as his son five years ago the sum of their ages was 30 years. Find the present ages.